

RESEARCH REPORT

# Beyond the Mammogram: Diagnostic Follow-Up and Equity in California Breast Cancer Control, 2014-2023

*An implementation report on the screening-to-diagnosis cascade, racial equity, dense breasts, and accountability measures*

**Evidence window** January 2014 through December 2023; current screening guidance used for interpretation

## Central conclusion

A mammogram produces public-health benefit only when the system converts an abnormal finding into a timely diagnosis and effective care. California should make diagnostic resolution, not screening completion alone, the central equity measure for community breast cancer programs.

## Executive summary

Community breast cancer programs frequently report the number of women contacted or mammograms completed. Those measures are necessary, but they do not establish that early detection occurred. An abnormal screening result must be communicated, evaluated with diagnostic imaging, resolved through biopsy when indicated, and connected to treatment. Delays or losses anywhere in that pathway can erase the expected benefit of screening.

The equity implications are substantial. The 2024 USPSTF recommendation reported breast cancer mortality of 27.6 per 100,000 among non-Hispanic Black women compared with 19.7 per 100,000 among non-Hispanic White women. It also reported triple-negative breast cancer incidence of 24.2 versus 12.3 per 100,000. Black women reported high mammography use, yet mortality remained markedly higher, underscoring that screening participation alone cannot explain or close the disparity [2].

Dense breasts create another implementation challenge. Nearly half of women have dense breasts, which can reduce mammographic sensitivity and modestly increase risk. The USPSTF found insufficient evidence to recommend for or against supplemental ultrasound or MRI after a negative mammogram solely on the basis of density. Community programs must therefore communicate uncertainty accurately while ensuring that clinically indicated diagnostic evaluation is accessible [2].

### Accountability principle

The most useful equity dashboard begins with the abnormal result. It asks who completed diagnostic evaluation, how long resolution took, where patients were lost, and whether the differences persist after stratification by race and ethnicity, language, insurance, geography, disability, and breast density.

## Key findings

- The screening-to-diagnosis cascade contains multiple handoffs, and each handoff can create inequitable delay.
- National Black-White mortality and triple-negative breast cancer differences are large despite high self-reported mammography use among Black women [2].
- The USPSTF emphasizes that screening benefit depends on equitable follow-up evaluation and treatment after abnormal findings [2].
- Nearly half of women have dense breasts; evidence remains insufficient to establish a universal supplemental-imaging strategy after a negative mammogram [2].
- Community programs should use closed-loop registries and named accountability for every abnormal result.
- Time-to-resolution measures should be reported as distributions and stratified equity outcomes, not only as a single overall average.

## 1. Background

Breast cancer screening is a process rather than a test. The process begins with identifying an eligible person and continues through scheduling, mammography, result communication, diagnostic evaluation, pathology, and treatment. A program can perform well at one step while failing at the next. For example, mobile screening can improve access to mammography but create a new gap if patients with abnormal findings must navigate an unfamiliar diagnostic system without support.

This process perspective is particularly important for equity. Communities facing structural barriers may have similar screening rates after years of outreach yet still experience longer diagnostic delays, lower access to high-

quality treatment, or more aggressive disease subtypes. Measuring only mammography can therefore create the appearance of equity while the consequential gap has shifted downstream.

## 2. Methods

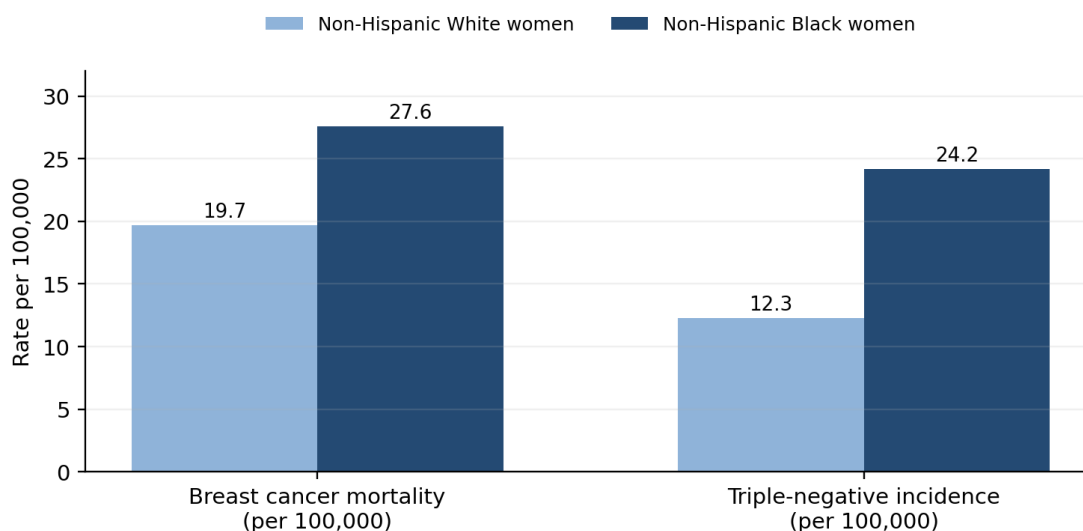
This report synthesized evidence and guidance relevant to breast cancer diagnostic follow-up during 2014–2023, with the 2024 USPSTF recommendation used to interpret current screening eligibility and equity priorities. Sources included federal recommendations, patient-navigation evidence, the California breast cancer prevention agenda, federal safety-net program documentation, and implementation-science frameworks.

The analysis mapped the breast cancer pathway into measurable stages and identified the operational requirements, failure modes, and equity indicators for each stage. Source-reported numerical values were retained without generating new population estimates.

## 3. Why screening rates are an incomplete equity measure

The USPSTF reported that Black women were approximately 40% more likely to die from breast cancer than White women. In the evidence summarized for the recommendation, mortality was 27.6 per 100,000 among non-Hispanic Black women and 19.7 per 100,000 among non-Hispanic White women. Triple-negative breast cancer incidence was 24.2 versus 12.3 per 100,000, respectively. At the same time, self-reported recent mammography among Black women was high. This pattern shows that equalizing screening is necessary but cannot by itself equalize outcomes [2].

The disparity can be produced by multiple mechanisms: differences in tumor subtype; delays in diagnostic resolution; access to specialists; treatment initiation and completion; comorbidities; quality of care; and social or financial conditions that shape the ability to remain in care. A community intervention should therefore identify the stage at which inequity appears rather than attributing all outcome differences to screening behavior.



**Figure 1. National Black-White differences in breast cancer mortality and triple-negative disease incidence**

Source: U.S. Preventive Services Task Force, 2024 [2]. Values are national benchmarks and should not be interpreted as California-specific rates.

## 4. The screening-to-diagnosis cascade

**Table 1. Breast cancer screening-to-diagnosis cascade and recommended accountability measures**

| Stage                         | Required action                                                                                    | Common failure mode                                                         | Recommended measure                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Eligibility and outreach      | Identify people due for risk-appropriate screening and communicate in a usable language and format | Incomplete registry data; inaccessible communication; outdated age criteria | Eligible population identified; successful contact rate; reach by subgroup |
| Scheduling and access         | Offer an appointment that is geographically, financially, and practically accessible               | Long waits; work conflicts; transportation; inaccessible equipment          | Days to appointment; completion; no-show rate; distance or travel burden   |
| Result communication          | Deliver understandable results promptly and document receipt                                       | Wrong contact information; language mismatch; portal-only communication     | Time to notification; confirmed receipt; preferred-language communication  |
| Diagnostic imaging            | Complete additional mammographic views and/or ultrasound when recommended                          | Separate facility; authorization delay; unavailable appointments            | Diagnostic imaging completion; days from abnormal screen                   |
| Biopsy and pathology          | Complete tissue diagnosis when indicated and communicate the result                                | Referral fragmentation; fear; cost; pathology delay                         | Biopsy completion; pathology turnaround; days to definitive diagnosis      |
| Treatment linkage             | Connect confirmed cancer to multidisciplinary care and initiate treatment                          | Insurance transition; specialty access; referral loss                       | Oncology consultation; treatment initiation; days from diagnosis           |
| Survivorship and surveillance | Coordinate ongoing surveillance, symptom management, and psychosocial support                      | Unclear responsibility; language barriers; financial toxicity               | Survivorship plan; surveillance completion; patient-reported unmet need    |

Measures should be stratified by race and ethnicity, language, insurance, geography, disability, age, and other locally relevant variables.

## 5. Dense breasts and communication of uncertainty

Nearly half of women have dense breasts. Dense tissue can make cancers more difficult to detect on mammography and is also associated with increased breast cancer risk. The 2024 USPSTF recommendation concluded that evidence was insufficient to determine the balance of benefits and harms of supplemental ultrasound or MRI after an otherwise negative screening mammogram solely because of breast density [2].

This is not a reason to avoid communication. It is a reason to communicate precisely. Community programs should explain what density means, distinguish routine screening from diagnostic evaluation, encourage discussion of individual risk, and avoid implying that every person with dense breasts requires the same additional test. Referral systems should support clinical risk assessment and shared decision-making without shifting the entire burden of interpretation to the patient.

### Communication standard

Density notification should answer four questions: What did the mammogram show? Does the patient need diagnostic follow-up now? How does density affect future screening? Who will help assess individual risk and decide whether additional imaging is appropriate?

## 6. Closed-loop follow-up as an intervention

A closed-loop system does not assume that ordering a test completes the task. It assigns responsibility, tracks status, escalates overdue cases, and documents resolution. For breast cancer programs, the loop begins when an abnormal result is generated and ends when the finding is resolved as benign, placed into a defined surveillance pathway, or confirmed as cancer with treatment linkage.

Navigation is one way to operationalize the loop. Meta-analytic evidence supports navigation for increasing cancer screening among populations affected by disparities, and the same barrier-resolution logic applies after abnormal

results [6]. The navigator should not substitute for clinical decision-making; the role is to ensure that the clinical decision can be completed despite logistical, financial, language, and organizational barriers.

## 7. A proposed California equity dashboard

**Table 2. Minimum breast diagnostic equity dashboard**

| Domain                | Core indicator                                                        | Preferred reporting format                              | Equity question                                                            |
|-----------------------|-----------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------|
| Reach                 | Proportion of the eligible population screened                        | Overall and subgroup percentages with denominators      | Who is not entering the pathway?                                           |
| Abnormal findings     | Rate of abnormal screens and documented notification                  | Counts, rates, and time-to-notification distribution    | Are result communication failures concentrated in particular groups?       |
| Diagnostic resolution | Proportion resolved within locally defined time thresholds            | Median, 25th/75th percentiles, and threshold completion | Who waits longest or remains unresolved?                                   |
| Biopsy                | Completion and time from recommendation to pathology                  | Completion proportion and time distribution             | Do authorization, facility, or language barriers affect biopsy completion? |
| Treatment linkage     | Oncology consultation and treatment initiation after diagnosis        | Proportion and time distribution                        | Does diagnostic equity translate into treatment equity?                    |
| Patient experience    | Reported understanding, trust, logistical burden, and unmet need      | Validated survey measures plus qualitative review       | Does the pathway work from the patient's perspective?                      |
| Implementation        | Navigator caseload, referral capacity, fidelity, and staff continuity | Site-level process metrics                              | Are outcome gaps driven by inadequate program capacity?                    |

*Programs should predefine numerators, denominators, exclusions, and the clinical meaning of time thresholds. Overall averages should never replace subgroup reporting.*

## 8. Program design implications

### 8.1 Assign ownership for unresolved results

Every abnormal result should have a named team or role responsible for tracking it until resolution. Responsibility should not disappear when a patient moves from primary care to radiology, from screening to diagnostics, or from a mobile unit to a hospital-based service.

### 8.2 Reserve diagnostic capacity before expanding screening volume

Screening campaigns can increase abnormal-result volume. Programs should estimate expected recalls and biopsies, establish referral agreements, and reserve diagnostic appointments before launching large outreach events. Otherwise, a successful screening campaign can lengthen the queue it depends on.

### 8.3 Build language access into every stage

Result letters, telephone communication, radiology preparation, consent, and oncology consultation must be available in the patient's preferred language. Community workers can reinforce understanding, but the health system remains responsible for qualified interpretation and accessible communication.

### 8.4 Treat data quality as an equity intervention

Incomplete race and ethnicity, language, insurance, and referral data make disparities invisible. Programs need standardized demographic fields, patient-preferred identity categories where feasible, and routine review of missingness. A dashboard that excludes people with incomplete data can falsely report improvement.

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## 9. Recommendations

1. Make diagnostic resolution and time to resolution required outcomes for publicly funded community breast cancer programs.
2. Create closed-loop registries that track every abnormal mammogram through resolution and treatment linkage.
3. Stratify all cascade measures by race and ethnicity, preferred language, insurance, geography, disability, age, and breast density where available.
4. Require diagnostic-capacity plans and referral agreements before funding large screening expansions or mobile events.
5. Provide navigation for patients with abnormal findings, prioritizing those with identified social or structural barriers.
6. Standardize breast-density communication and connect notification to individual risk assessment and shared decision-making.
7. Report distributions and overdue cases rather than relying on overall mean time to follow-up.
8. Use patient experience and implementation data to explain where and why the pathway fails.

## 10. Limitations

This report does not estimate California-specific diagnostic-delay rates because publicly available data are not consistently reported across health systems and counties using comparable definitions. National racial-disparity values are used as benchmarks and are clearly labeled. The proposed dashboard requires local validation of clinically appropriate time thresholds, data availability, and workflow ownership.

## 11. Conclusion

California's breast cancer equity agenda should move beyond counting mammograms. The central question is whether every abnormal finding becomes a timely, understandable, and completed diagnostic pathway. Programs that measure and finance that pathway can identify downstream inequities that screening rates conceal and can protect the benefit that early detection is intended to deliver.

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